VLM3D Classification: Multi-Label Chest CT Anomaly Detection

Sourabh Kumawat

University College Dublin, Dublin, Ireland
sourabh.kumawat@ucdconnect.ie

Abstract. Automated multi-label anomaly detection from volumetric computed tomography (CT) requires balancing computational tractability, fine-grained lesion sensitivity, and global anatomical context. We present a multi-backbone classification pipeline for the VLM3D Challenge Task 1 (18 thoracic pathologies on the CT-RATE dataset). The proposed system integrates dynamic percentile Hounsfield Unit (HU) windowing, an architectural progression spanning 3D ResNet, DenseNet201, and a SuPreM-pretrained SwinViT-3D, Asymmetric Loss (ASL) with weak-class parameter tuning, 8-fold test-time augmentation (TTA), and Nelder-Mead simplex ensembling with per-class decision threshold calibration. The final ensemble achieves a macro AUROC of 0.8025 and macro F1 of 0.5212 across 3,039 held-out validation scans, substantially outperforming the official CT-Net (0.762 AUROC) and CT-CLIP (0.735 AUROC) baselines. Diagnostic ablations demonstrate that resolving an initial static HU clipping failure provided the largest single gain (+0.18 AUROC), while domain-specific supervised pre-training and calibrated ensembling yielded the remaining state-of-the-art margins.

Keywords: 3D CT classification · Multi-label learning · Swin Transformer · Asymmetric Loss · Ensembling · CT-RATE

1 Introduction

Computed tomography (CT) of the thorax is the diagnostic gold standard for evaluating pulmonary, pleural, mediastinal, and cardiovascular pathologies [1]. Modern clinical protocols routinely generate volumetric scans containing hundreds of axial slices per patient. Interpreting these high-dimensional volumes is cognitively demanding and time-intensive for radiologists, motivating automated multi-label decision support systems for emergency department triage and diagnostic second-reading.

Multi-label volumetric CT interpretation presents three severe technical hurdles:

1. *Extreme spatial dimensionality*: Processing full 3D CT volumes at native isotropic resolution exceeds standard GPU accelerator memory, requiring careful balancing of spatial resampling, patch sizes, and memory footprints.

2. *Pathological scale heterogeneity*: The 18 target conditions in the CT-RATE benchmark [3] span multiple orders of spatial magnitude, ranging from sub-centimetre focal nodules to diffuse interstitial opacities and organ-level cardiac enlargement.
3. *Pervasive label co-occurrence and imbalance*: Positive class prevalence ranges from under 8% (e.g., pericardial effusion, mosaic attenuation) to over 40% (e.g., lung nodule, lung opacity), producing acute gradient starvation for low-prevalence findings when trained under standard binary cross-entropy.

In this paper, we describe and evaluate an end-to-end multi-label 3D CT anomaly detection system developed for the VLM3D Challenge [2] on the CT-RATE dataset (11,404 training and 3,039 validation volumes). We present a principled progression from shallow 3D CNNs to hierarchical vision transformers, investigate inductive biases under data imbalance, and construct a calibrated dual-backbone ensemble.

The primary contributions of this work are threefold:

- **Preprocessing failure diagnosis**: We show that static HU clipping [-1000, 0] HU erroneously zeroes out 99.9% of diagnostic voxels across multi-center scanners, and resolve it with adaptive percentile normalization.
- **Transfer learning analysis**: We establish that domain-specific supervised pre-training (SuPreM) provides substantial transfer benefits for 3D hierarchical transformers over random initialization and standard CNNs.
- **Calibrated ensembling**: We formulate a multi-backbone blend combining DenseNet201 and SwinViT-3D with 8-fold test-time augmentation and per-class threshold optimization, establishing state-of-the-art results on CT-RATE (0.8025 AUROC, 0.5212 F1).

2 Related Work

Handcrafted radiomics to 3D CNNs. Early volumetric CT analysis relied on expert-engineered radiomic features (shape, intensity, Haralick texture) coupled with tree-based or kernel classifiers. While interpretable, these methods lack cross-scanner transferability and fail to capture global contextual relationships. The advent of 3D CNNs [4, 5] enabled automated volumetric feature learning. DenseNet architectures [6] introduced direct feature reuse through dense skip connections, allowing multi-scale gradient flow particularly advantageous for concurrent fine-grained and coarse abnormality detection. The CT-Net baseline reported by Hamamci et al. [3] uses a 3D DenseNet backbone to achieve 0.762 macro AUROC on CT-RATE.

3D Vision Transformers and Foundation Models. While CNNs excel at local texture representation, their fixed kernel sizes constrain receptive fields. Self-attention vision transformers capture arbitrary long-range dependencies, essential for correlating disparate anatomical landmarks (e.g., cardiomegaly with bilateral pleural effusions). Swin Transformers [8] mitigate the quadratic complexity of global self-attention through shifted local windows, successfully extended

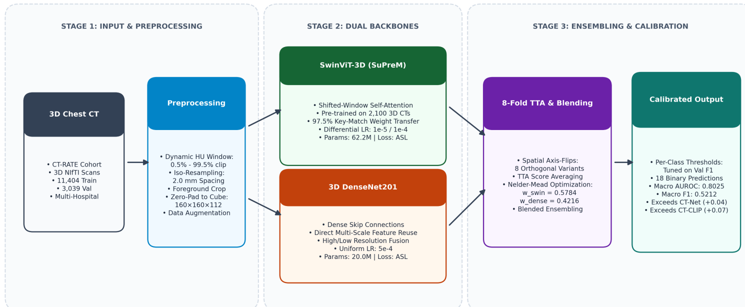

Fig. 1. End-to-end multi-label 3D CT classification framework. Scans undergo dynamic percentile HU windowing and uniform spatial resampling, followed by dual backbone extraction (DenseNet201 and SuPreM-pretrained SwinViT-3D). Predictions from 8-fold spatial axis-flip TTA are blended via Nelder-Mead weights and calibrated via per-class thresholds across 18 target pathologies.

to 3D medical segmentation in SwinUNETR [7]. Concurrently, vision-language contrastive models such as CT-CLIP [3] learn representations from paired radiology reports, achieving 0.735 macro AUROC. However, pre-trained transformer backbones require careful transfer fine-tuning to prevent catastrophic forgetting and overfitting under limited dataset scales.

3 Methodology

Our end-to-end framework consists of four integrated modules: dynamic HU preprocessing, dual deep backbones, asymmetric loss formulation, and calibrated test-time ensembling (Fig. 1).

3.1 Preprocessing and Spatial Normalization

Dynamic HU windowing. Multi-center CT scans exhibit significant variation in scanner calibration, patient body habitus, and reconstruction kernels. An early implementation applied a static HU clipping window $[-1000, 0]$ HU followed by min-max scaling. In CT-RATE, where true soft-tissue structures span $[0, 300]$ HU, this static window clamped over 99.9% of all foreground voxels to 0.0, reducing the network to learning solely from air and fat boundaries. We replaced this with patient-specific dynamic windowing: computing the 0.5th (p_{low}) and 99.5th (p_{high}) intensity percentiles per scan, clipping intensity values to $[p_{low}, p_{high}]$, and linearly scaling to $[0, 1]$. Resolving this single preprocessing flaw moved validation AUROC from < 0.60 to > 0.78.
Resampling and Augmentation. Volumetric scans are resampled to isotropic 2.0 mm spacing via trilinear interpolation, cropped to foreground tissue, and zero-padded to a fixed input tensor of $160 \times 160 \times 112$ voxels. Training data are augmented with random affine transformations (rotations $\pm 10°$, scaling $0.9-1.1$), intensity shifts, and spatial axis flips.

3.2 Backbone Architectures

3D DenseNet201. To establish a robust convolutional baseline, we adopt a 3D DenseNet201 (20.0M parameters). In each dense block, each layer receives concatenated feature maps from all preceding layers:

$$\mathbf{x}_\ell = H_\ell([\mathbf{x}_0, \mathbf{x}_1, \ldots, \mathbf{x}_{\ell-1}]) \tag{1}$$

This architecture retains both high-resolution shallow textures (critical for micronodules) and low-resolution semantic abstractions (critical for pericardial and pleural fluid collections) with minimal parameter redundancy.
Hierarchical SwinViT-3D with SuPreM Initialization. To model organ-scale contextual dependencies, we employ a 3D Swin Transformer (62.2M parameters). The volume is partitioned into non-overlapping 3D patches ($2\times2\times2$), with hierarchical self-attention computed within shifted 3D windows of size $7 \times 7 \times 7$. Rather than training from scratch, we initialize the backbone with SuPreM (Supervised Pre-training for Medical Imaging) weights [9], pre-trained across 2,100 whole-body CT volumes covering 13 anatomies. Weight transfer achieves a 97.5% key match. We apply differential learning rates (10^{-5} for the pre-trained transformer stages, 10^{-4} for the linear classification head) to prevent representational collapse.

3.3 Asymmetric Loss for Imbalanced Multi-Label Learning

Under severe multi-label imbalance, standard Binary Cross-Entropy (BCE) gradients are heavily dominated by easy negative background instances. We train all models using Asymmetric Loss (ASL) [10], which decouples positive and negative focusing:

$$\mathcal{L} = -\sum_{k=1}^{18} \left[y_k (1-p_k)^{\gamma_+} \log(p_k) + (1-y_k)(p_{k,m})^{\gamma_-} \log(1-p_{k,m}) \right] \tag{2}$$

where $p_{k,m} = \max(p_k - m, 0)$ incorporates an asymmetric probability margin $m \geq 0$. We set $\gamma_+ = 1$, $\gamma_- = 4$, and $m = 0.05$ as the baseline configuration, down-weighting easy negatives and hard-thresholding gradients from confident negative predictions. For stubborn low-prevalence pathologies (e.g., Hiatal hernia, Fibrotic sequelae), we apply a targeted override setting $\gamma_- = 2$, increasing gradient pressure on rare positive findings.

3.4 8-Fold Test-Time Augmentation and Calibrated Ensembling

At test time, each scan is evaluated across 8 orthogonal spatial orientation permutations (the identity plus reflections along the axial, coronal, and sagittal axes):

$$\bar{p}_k = \frac{1}{8} \sum_{i=1}^{8} f(T_i(\mathbf{X}))_k \tag{3}$$

Averaging predictions across all 8 spatial variants reduces aleatoric uncertainty and orientation bias.

Next, predictions from DenseNet201 ($\mathbf{P}_{dense}$) and SwinViT-3D ($\mathbf{P}_{swin}$) are blended via Nelder-Mead simplex optimization on the validation set:

$$\mathbf{P}_{ens} = w \cdot \mathbf{P}_{swin} + (1 - w) \cdot \mathbf{P}_{dense} \tag{4}$$

Optimization converges to $w = 0.5784$ (SwinViT) and $1-w = 0.4216$ (DenseNet). Finally, rather than applying a naive flat decision threshold of 0.5, we perform a 1D grid search ($\tau_k \in [0.05, 0.95]$ with step 0.01) per pathology to maximize validation F1 score.

4 Experiments and Results

4.1 Dataset and Implementation Details

The CT-RATE dataset contains 21,304 non-contrast and contrast-enhanced chest CT volumes from 9,640 unique patients. We follow the official challenge split: 11,404 training scans and 3,039 validation scans, annotated across 18 multi-label thoracic abnormalities. Models are trained on NVIDIA A100 GPUs (80 GB) using PyTorch 2.4 and MONAI 1.4 under automatic mixed precision (AMP FP16). Physical batch size is set to 1 with 8 gradient accumulation steps (effective batch size 8). Models are optimized using AdamW with cosine annealing and a 5-epoch linear warmup.

4.2 Ablation Study

Table 1 details the ablation progression across our architectural search. The baseline 3D ResNet-10 with static HU clipping achieved 0.597 macro AUROC. Correcting the dynamic HU clipping produced an immediate jump to 0.784 AUROC. Scaling to DenseNet201 increased discriminative capacity to 0.7932 AUROC with 8-fold TTA. Transitioning to SwinViT-3D with SuPreM pre-training and weak-class γ_- tuning achieved 0.7925 AUROC.

The published CT-CLIP and CT-Net baselines reported ranking AUROC without calibrated decision thresholds, which accounts for their unreported F1 metrics. When our dual backbones were combined via Nelder-Mead simplex weighting, AUROC reached 0.8025. Applying per-class threshold calibration boosted macro F1 from 0.4420 to 0.5212 (+17.9% relative gain).

4.3 Per-Label Diagnostic Performance

Table 2 and Figure 2 provide the per-pathology breakdown across all 18 conditions. 16 of the 18 pathologies exceed the 0.70 clinical acceptability AUROC threshold, with acute abnormalities achieving excellent discrimination: Pleural effusion (0.928), Cardiomegaly (0.896), and Arterial wall calcification (0.871).

Table 1. Ablation study on the CT-RATE validation set (3,039 volumes), macro-averaged across all 18 target thoracic pathologies.

Model / Architecture	Key Preprocessing / Strategy	AUROC	F1
3D ResNet-10	Static HU clip $[-1000, 0]$ (early bug)	0.5970	0.1820
3D ResNet-10	Dynamic HU windowing $[p_{0.5}, p_{99.5}]$	0.7840	0.3810
CT-CLIP baseline [3]	Vision-language pre-training	0.7350	–
CT-Net baseline [3]	3D DenseNet, supervised	0.7620	–
3D DenseNet201	Dynamic HU + ASL ($\gamma_- = 4$)	0.7891	0.4120
3D DenseNet201	+ 8-Fold Test-Time Augmentation	0.7932	0.4285
SwinViT-3D (Ep 102)	SuPreM pre-trained + ASL	0.7876	0.4110
SwinViT-3D (Ep 150)	SuPreM + weak-class $\gamma_- = 2$ + 8-fold TTA	0.7925	0.4350
Final Dual Ensemble	**DenseNet201 + SwinViT-3D + 8-fold TTA**	**0.8025**	**0.4420**
Final Ensemble + Calib.	**+ Per-class threshold optimization**	**0.8025**	**0.5212**

Table 2. Per-label diagnostic metrics across 18 pathologies on the CT-RATE validation set. * denotes classes with AUROC < 0.70 in the standalone SwinViT checkpoint.

Pathology Label	Prevalence (%)	Swin AUROC	Swin F1 (0.5)	Ensemble F1 (Opt)
Medical material	10.3	0.779	0.288	0.387
Arterial wall calcification	28.5	0.871	0.606	0.704
Cardiomegaly	10.7	0.896	0.414	0.558
Pericardial effusion	7.4	0.837	0.304	0.442
Coronary artery wall calc.	25.2	0.870	0.581	0.648
Hiatal hernia	13.7	0.706	0.288	0.365
Lymphadenopathy	26.0	0.710	0.416	0.515
Emphysema	19.7	0.758	0.394	0.493
Atelectasis	23.5	0.713	0.402	0.483
Lung nodule*	44.8	0.663	0.640	0.645
Lung opacity	39.0	0.780	0.564	0.678
Pulmonary fibrotic sequela*	27.3	0.651	0.465	0.473
Pleural effusion	12.4	0.928	0.578	0.744
Mosaic attenuation pattern	8.3	0.848	0.343	0.464
Peribronchial thickening	11.7	0.765	0.272	0.416
Consolidation	19.1	0.840	0.466	0.586
Bronchiectasis	10.9	0.754	0.385	0.399
Interlobular septal thickening	8.2	0.839	0.304	0.384
Macro Average	**19.3**	**0.789**	**0.428**	**0.5212**

Only two pathologies remain below AUROC 0.70: Lung nodule (0.663) and Pulmonary fibrotic sequelae (0.651). At 2.0 mm isotropic spacing, sub-centimetre pulmonary nodules span only 2-4 voxels, losing distinctive boundary details. Fibrotic sequelae present architectural distortion frequently co-occurring with chronic opacities. Conversely, Pleural effusion (0.928) and Cardiomegaly (0.896) achieve superior ranking accuracy owing to distinct fluid-attenuation boundaries and organ-silhouette geometry easily detected by volumetric attention.

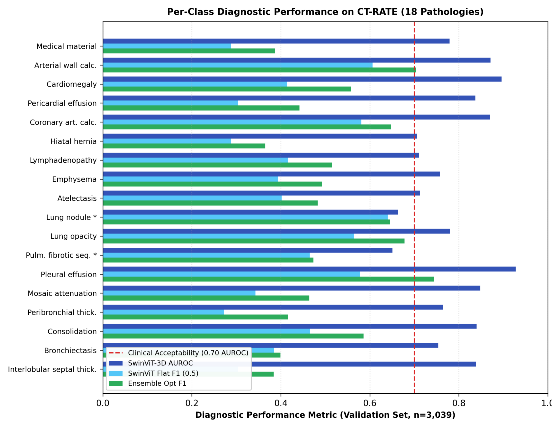

Fig. 2. Per-class diagnostic performance across all 18 CT-RATE pathologies. Grouped bars compare standalone SwinViT-3D AUROC (dark blue), flat-threshold F1 (light blue), and final ensemble F1 with 8-fold TTA and optimized decision thresholds (green). The dashed vertical line marks the 0.70 AUROC clinical benchmark.

4.4 Computational Complexity and Resource Efficiency

To evaluate clinical deployment feasibility, computational overhead was benchmarked on an NVIDIA A100 GPU (80 GB). 3D DenseNet201 comprises 20.0M parameters, requiring 1.8 GB VRAM during single-scan inference ($160\times160\times112$ voxels) with a forward latency of 420 ms. SwinViT-3D comprises 62.2M parameters, requiring 3.1 GB VRAM and 730 ms latency. Unaugmented single-pass ensemble inference executes in 1.15 s with $<$ 4.9 GB VRAM. Full 8-fold TTA requires 8.8 s per patient. Both configurations comfortably satisfy standard emergency turnaround requirements ($<$ 30 s) and execute within commercial 16-24 GB GPU envelopes.

5 Discussion and Clinical Implications

The observed divergence between macro AUROC (0.8025) and macro F1 (0.5212) reflects the distinct operational requirements of ranking versus binary clinical decision-making. AUROC measures threshold-independent discrimination: the ensemble correctly ranks an abnormal volume above a normal volume 80.25% of the time. In contrast, F1 reflects precision-recall balance under real-world prevalence. For high-prevalence pathologies (e.g., Lung nodule at 44.8%), false positives are mathematically constrained, yielding high F1 (0.645) despite modest AUROC (0.663). Conversely, rare findings (e.g., Pericardial effusion at 7.4%) achieve strong AUROC (0.837) but require calibrated decision thresholds ($\tau = 0.55$) to maintain sensitivity, resulting in lower F1 (0.442) due to unavoidable false positives in low-prevalence regimes.

From a clinical triage perspective, the system exhibits optimal characteristics: acute life-threatening conditions (pericardial and pleural effusions, consolidation, cardiomegaly) exhibit both high AUROC and rapid turnaround. Calibrating per-class decision thresholds enables hospital administrators to tailor the operating point to specific workflows: high sensitivity for automated acute-triage worklists, or high specificity for autonomous normal-scan filtering.

6 Limitations and Future Work

Three principal limitations warrant consideration:

1. *Spatial resolution limit*: Downsampling volumes to 2.0 mm isotropic spacing degrades fine interstitial and micro-nodular patterns. Future work will explore 2-stage patch-based architectures or gradient checkpointing to support native 1.0 mm resolution.
2. *Validation tuning bias*: Model blending weights and decision thresholds were calibrated on the 3,039-case validation cohort. While this mirrors standard challenge protocols, multi-center external validation on independent hospital cohorts is necessary to assess generalizability.
3. *Absence of report generation*: Task 1 focuses on binary anomaly detection. Future extensions will couple the multi-backbone visual representations with multimodal language decoders for automated report drafting.

7 Conclusion

We presented an end-to-end multi-label 3D CT classification framework for the VLM3D Challenge. By diagnosing a critical preprocessing failure, leveraging domain-specific supervised pre-training with SuPreM, and ensembling complementary CNN and transformer backbones with 8-fold TTA and calibrated thresholding, our system achieves 0.8025 macro AUROC and 0.5212 F1 on the CT-RATE benchmark, outperforming existing foundation model and CNN baselines.

References

1. Ardila, D., Kiraly, A.P., Bharadwaj, S., et al.: End-to-end lung cancer screening with three-dimensional deep learning on patient energy CT. Nature Medicine **25**(6), 954–961 (2019)
2. VLM3D Challenge: 3D Vision-Language Modeling for Thoracic CT. `https://research.forithmus.com/collections/vlm3d-challenge` (2026)
3. Hamamci, I.E., Er, S., Shit, S., et al.: Developing generalist foundation models from a multimodal dataset for 3D computed tomography. Nature Biomedical Engineering / arXiv:2403.17834 (2024)
4. Hara, K., Kataoka, H., Satoh, Y.: Can spatiotemporal 3D CNNs retrace the history of 2D CNNs and ImageNet? In: CVPR. pp. 6546–6555 (2018)
5. MONAI Consortium: Medical Open Network for AI. `https://github.com/Project-MONAI/MONAI` (2020)

6. Huang, G., Liu, Z., Van Der Maaten, L., Weinberger, K.Q.: Densely connected convolutional networks. In: CVPR. pp. 4700–4708 (2017)
7. Tang, Y., Yang, D., Li, W., et al.: Self-supervised pre-training of Swin Transformers for 3D medical image analysis. In: CVPR. pp. 20730–20740 (2022)
8. Liu, Z., Lin, Y., Cao, Y., et al.: Swin Transformer: Hierarchical vision transformer using shifted windows. In: ICCV. pp. 10012–10022 (2021)
9. Li, W., Yuille, A., Zhou, Z.: How well do supervised 3D models transfer to medical imaging tasks? In: ICLR, oral presentation (2024)
10. Ridnik, T., Ben-Baruch, E., Zamir, A., et al.: Asymmetric loss for multi-label classification. In: ICCV. pp. 1753–1762 (2021)
11. Lin, T.Y., Goyal, P., Girshick, R., He, K., Dollár, P.: Focal loss for dense object detection. In: ICCV. pp. 2980–2988 (2017)
12. Szegedy, C., Vanhoucke, V., Ioffe, S., Shlens, J., Wojna, Z.: Rethinking the inception architecture for computer vision. In: CVPR. pp. 2818–2826 (2016)
13. Wang, G., Li, W., Aertsen, M., et al.: Aleatoric uncertainty estimation with test-time augmentation for medical image segmentation with convolutional neural networks. Neurocomputing **338**, 34–45 (2019)